Lecture 1

Monday, 16 January 2023 7:22 AM

→ Electrochem Engineering → Translating fundamental principles to applied/industrial contexts
↳ (of electrochem transformations)

→ Why?
↦ CO_2-free (green e^-) → Decarbonization
↦ Driven by e^- rather than heat

→ How?
↦ Concepts: Thermo, kinetics, Transport
Charge, voltage, conservation of charge } Fundamentals

↦ Current distributions, Rotating electrodes
Electroanalytical methods } Techniques, modelling to test and understand e-chem systems

↦ Porous electrodes/batteries,
Electrocatalysis, decarbonization } Applied systems

→ Goal?
↦ Technical working knowledge of e-chem phenomena for e^- driven processes

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Electrochemical reaction: A change in the molecular identity of a chemical species in which one or more e^- are reactant or product

product

eg: $2\,Cl^-_{(aq)} \rightarrow Cl_{2\,(g)} + 2e^-$

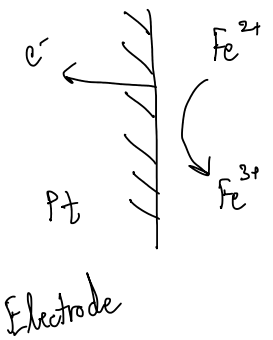
(rxn does not occur unless e^- transfer happens)

↳ e^- transfer to/from electrode to a different phase

↓ ↓

have to be free e^- — conducting/semi-conducting

↳ echem reactions are inherently heterogeneous.



- Driving force: Thermodynamics (e-chem potential)
- Proximity: Physics
- Transport: interfacial region

↳ Charge neutrality: takes an enormous amount of energy to separate charge. Therefore, we can assume charge neutrality everywhere except over very short distances.

↓

This means that there must be another electrode

↳ Charge transfer @ each electrode ("half reaction")

↳ Half reactions are chemically independent.